Technical Report – Airbnb Rental Prices

In this project, we are attempting to predict Airbnb rental prices from 16 variables, including number of bedrooms, number of reviews, and whether the host is a superhost. We are provided with a training set of 3208 observations and a test set of 700 observations, the latter of which we are going to use to create predictions from.

Exploratory Data Analysis

Univariate analysis

We do not observe any missing values or apparent data entry errors in this dataset. However, one of the notable aspects of this dataset is the number of influential points. Indeed, we find that 250 of the 3028 points – over 8% of the observations – are counted as outliers in the distribution of price. Additionally, we find that 95 observations in the training dataset are considered influential points by the threshold of 4/n for Cook's distance. These results are partially expected given that price is right-skewed, but the number of outliers suggests that one of the primary concerns with this dataset will be handling these extreme values.

Looking at the plots of the predictors, we do not notice any greatly unusual values. Many predictors have distributions that are right-skewed, such as bathrooms, bedrooms, accommodates, and number of host listings. However, some predictors are also left-skewed, such as number of days available and review scores. It is important to note that the dataset contains two categorical predictors: neighborhoods and room type, which will need to be transformed into numeric data for fitting decision trees (and therefore also random forests).

Bivariate and multivariate analysis

Accomodates is relatively correlated with room type (-0.45), bathrooms (0.64), and bedrooms (0.77). Review scores rating is also closely correlated with review scores cleanliness

(0.74) and review scores location (0.59). However, VIF does not indicate collinearity between variables, as they are not closely related enough to be of concern.

We see a number of influential points – 95 by the 4/n threshold – by the plot of Cook's distance. However, four appear especially different: observation 635 with a distance of 0.2937, 2139 with 0.1868, 1596 with 0.1371, and observation 337 with a Cook's distance of 0.1056. Looking at the values of the predictors for these four observations, we do not see any particularly unusual values, meaning that the prices listed are outliers and do not add significant information to the models.

However, we still want to be cautious about deleting valid data points simply because they do not fit the model. Therefore, we will fit models with both the full dataset and the four most influential points deleted.

Model Analysis

<u>Models with all predictors</u>

The full linear model results in a training mean squared error (MSE) of about 28314. This means that, on average, each prediction is about sqrt(28314) = $168 off from the true price of the rental. Fitting the full model with the four most influential points deleted, we find a root mean squared error (RMSE) of about $122, a notable improvement from the previous model.

However, since the response variable – price – is right-skewed, we can log transform the response variable to offset the lack of normality. When we fit a generalized linear model (GLM) with a log link, we see a slightly improved RMSE of about $163. The GLM with the deleted observations has a RMSE of about $118.

Given the apparent lack of collinearity between predictors by checking the VIF, we expect that ridge and lasso regression should not perform significantly better than ordinary least squares regression – the decrease in variance does not offset the increase in bias when using regularization. This is shown to be the case when we fit the full ridge and lasso models with the best lambda value selected through cross validation; they both perform nearly identically to the OLS regression model, with a RMSE of about $168. The ridge and lasso models fitted on the dataset with the deleted observations have RMSEs of about $122, again similar to the OLS regression model with the deleted observations.

At this point, the model with the lowest RMSE is the generalized linear model with the deleted observations. This makes sense given the fact observed that there is not significant variance in the dataset that would make regularization through ridge or lasso regression especially better. On the other hand, the right-skewed nature of the response variable makes a log link for the GLM beneficial.

Fitting an unpruned decision tree, we find a RMSE of about $165 for the full dataset and a RMS of about $118 for the dataset with deleted observations. While this performs similarly to the GLM with the log link, we want to see if there is an improvement with a pruned tree. We find that the tree has the minimum deviance with 6 terminal nodes, but this tree provides a RMSE of $170, the highest error of all models tested so far, by a small margin.

We already expect the random forest model to perform the best of the models tested. Given the correlation observed between some predictors, random forest should be able to offset this collinearity because of the random subset of predictors it selects for each tree in the forest. Indeed, when we fit the random forest model, we observe the smallest RMSEs by far: slightly over $81 for the full dataset and $56 for the dataset with deleted observations.

<u>Models with subset of predictors</u>

At this point, we expect the random forest model to be the ideal model. However, to check the effect of the many predictors, we can check the most important predictors using forward selection. Fitting the linear models, GLMs, and ridge and lasso models with the most important predictors taken from forward selection, we – as expected – do not see a significant difference in RMSE between the full models and the partial models. Hence, our best model appears to be the random forest model.

Summary of Results

Models	**RMSE (full/partial)**
Linear model	168.27/168.29
Linear model (-4)	*122.59/122.59*
GLM with log link	162.93/163.43
GLM with log link (-4)	*118.31/118.36*
Ridge	168.42/168.44
Ridge (-4)	*122.8/122.8*
Lasso	168.41/168.43
Lasso (-4)	*122.92/122.86*
Decision tree	164.68
Decision tree (-4)	*118.23*
Pruned tree	169.68
Random forest	**81.44**
Random forest (-4)	***56.01***

It is noticeable that every partial model performed almost identically to the full models. This shows that several of the predictors, such as number of host listings or minimum nights were not important in explaining price.

Conclusions and Takeaways

As mentioned earlier, we expected random forest to perform the best of the models fitted. In general, random forest is robust to outliers, multicollinearity, and overfitting. Although the data is not collinear by the variance inflation factor, we observed some correlated predictors in the correlation matrix. Additionally, because we are using training RMSE as our measure of accuracy, overfitting is a concern, and the many outliers in the response variable suggest a more complicated relationship than the dataset might capture. However, random forest does the best job of offsetting these concerns.

It is also notable that – apart from random forest – the majority of the models that we fit performed similarly despite their differing strengths.

Given the many outliers in the training dataset, it seems reasonable to assume that the test dataset could have similar observations that have extreme response values that would not be predictable from the values of the features. Therefore, while every model that was fit on the dataset without the four most influential points has a significantly lower RMSE than its full-dataset counterpart, this likely is not representative of what the test RMSE would be. Therefore, assuming the test set is similar to the training set, we could expect the test RMSE to be similar to the training RMSE for the full dataset.

In the future, it could be beneficial to either record more observations, offsetting the impact of the influential points. However, this could still result in a similar amount of influential points in the new data. A potentially better approach in the future could be to measure the values of other potential predictors to investigate if any explain the presence of the outliers in the dataset.